

The Capacity of Black Holes to Transmit Quantum Information [1]: A Review

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(Dated: May 17, 2026)

We review the quantum information-theoretic treatment of black holes developed by Bradler and Adami [1], wherein a Schwarzschild black hole in the semiclassical limit is modelled as a quantum channel and its *quantum capacity* is computed in two exactly solvable limits. In the perfectly reflecting limit the channel is isomorphic to the Unruh channel and carries a non-vanishing quantum capacity, so that an outside observer can in principle reconstruct Alice’s entanglement from stimulated emission. The complementary interior channel is entanglement-breaking, guaranteeing that the no-cloning theorem is never violated. In the perfectly absorbing limit the roles are exchanged: the exterior channel becomes entanglement-breaking with vanishing capacity, while the interior channel acquires a non-vanishing capacity. In both cases the no-cloning theorem is respected, dissolving the apparent “information paradox” without invoking new physics beyond quantum Shannon theory. We also contextualise some connections to quantum chaos and information scrambling as promising future directions.

I. INTRODUCTION

Hawking’s 1975 discovery that black holes emit thermal radiation [2] to evaporate initiated one of the most enduring puzzles in theoretical physics: the *black hole information paradox*. Hawking radiation is perfectly thermal and appears to carry no imprint of whatever quantum state fell into the black hole, suggesting that a pure state can evolve into a mixed one. This very breakdown of unitarity was once considered the most paradoxical aspect of the problem, but it is well known now that in the context of quantum Shannon theory, a pure state can very naturally evolve into a mixed state if part of a larger, pure system is traced over.

Thus, it is probably not an incomplete understanding of quantum gravity that has hindered progress, but a precise formulation of the problem within the natural framework of quantum information that has been missing. The apparent paradox thus could be solved if the correct question is asked: not whether the final state is mixed, but whether the *entanglement* that a sender, Alice, possessed before sending her quantum message into the black hole can be faithfully reconstructed by a distant observer, Bob. The transmission of this entanglement is precisely the problem quantum information theory addresses.

Naturally, one may first begin by asking classical information theoretic questions. The work by Adami and ver Steeg [3] showed that the *classical* capacity of a black hole (the Holevo capacity) is positive. Classical information sent into the black hole does *not* escape in Hawking radiation; rather, it reappears in the *stimulated* emission that any absorber must produce in response to incoming quanta [8]. This resolves the classical version of the paradox without invoking new physics. We will later show how Hawking radiation cannot even carry quantum information, and is thus totally *featureless*.

The question of *quantum* information is harder. Hay-

den and Preskill [4] were the first to frame it in the language of quantum Shannon theory. They modelled the black hole as a random unitary “scrambler” and argued that, once a black hole has radiated more than half its initial entropy (the Page time [5]), any new qubit thrown in is rapidly re-emitted in the Hawking radiation alongside a small number of early-emitted qubits, provided the observer possesses a large enough fraction of the early radiation. This picture suggests that black holes are remarkably efficient information mirrors, but it relied on a finite-dimensional, random-unitary model that does not obviously extend to the infinite-dimensional bosonic Hilbert spaces of real scalar fields around a Schwarzschild black hole. It should be noted that the authors label the all the output of the abstract black hole unitary as “Hawking radiation”, but in the context of this paper we distinguish Hawking radiation (spontaneous emission) from stimulated emission, both of which escape the black hole.

More recently, Yoshida and Kitaev [6] and Brown et al. [7] provided explicit *decoding* protocols for the Hayden-Preskill scenario, connecting the problem to quantum error correction and the structure of approximate quantum error-correcting codes. These protocols work efficiently when the scrambling unitary has high *quantum complexity*.

The paper by Bradler and Adami [1], which we review here, takes a complementary approach: rather than modelling the black hole as an abstract scrambler, they employ Sorkin’s consistent semiclassical formulation [8] to derive the explicit channel isometry from first principles, and then compute the quantum capacity directly for two limiting cases: perfectly reflecting and perfectly absorbing black holes. This is because the quantum capacity is tractable to calculate in these two limits, where the channel is provably single-letter.

II. FRAMEWORK

A. Describing the Scenario

We model the black hole as an open quantum system in the Schwarzschild geometry, treating it as a fixed background quantum channel in the semiclassical limit. The standard scenario (see Fig. 1) involves two observers: Alice, who sends a quantum state into the black hole from outside the horizon, and Bob, who collects the outgoing radiation at future null infinity $\mathcal{I}^+$. Alice's goal is to transmit her *entanglement* with a reference system R to Bob, so that after the interaction Bob is entangled with R in exactly the same way Alice was. This operational definition makes the problem of quantum information transmission precise.

The evolution of the combined system (black hole + exterior fields + interior fields) is perfectly unitary, but Bob can only access a subset of the degrees of freedom. Tracing over the inaccessible interior gives rise to a *quantum channel* $\mathcal{N} : \phi_A \mapsto \sigma_B$, a completely positive, trace-preserving (CPTP) map.

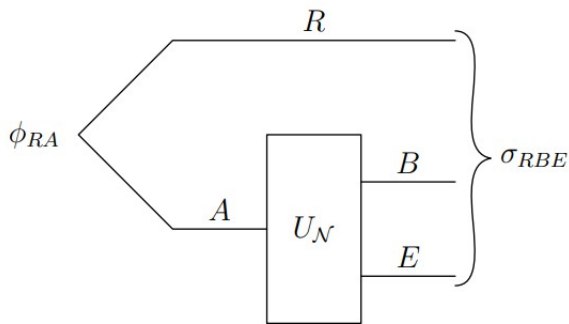


FIG. 1: Alice's state is unitarily transformed by U_N . The channel $\mathcal{N} : A \rightarrow B$ is obtained by tracing out RE . Adapted from Ref. [1].

B. Defining Quantum Capacity

The *quantum capacity* $Q(\mathcal{N})$ quantifies the maximum rate at which Alice can transmit quantum information (entanglement) reliably through the channel. Formally, if the channel is “not too noisy”, there exist an encoder $\mathcal{E}$ and decoder $\mathcal{D}$ such that [10, 11]

$$\|\mathcal{D} \circ \mathcal{N}^{\otimes n} \circ \mathcal{E} - \text{id}\|_{\diamond} \leq \varepsilon \quad (2.1)$$

for arbitrarily small ε as $n \rightarrow \infty$ and where $\|\cdot\|_{\diamond}$ is the diamond norm. The capacity $Q(\mathcal{N})$ is given as

$$Q(\mathcal{N}) = \lim_{n \rightarrow \infty} \frac{1}{n} Q^{(1)}(\mathcal{N}^{\otimes n}) \quad (2.2)$$

$$Q^{(1)}(\mathcal{N}) = \max_{\phi_{RA}} [H(B)_{\sigma} - H(E)_{\sigma}], \quad (2.3)$$

where $H(S)_{\sigma}$ denotes the von Neumann entropy of subsystem S of the output state $\sigma_{RBE} = (\text{id} \otimes U_N)(\phi_{RA})$, and the maximisation is over all entangled input states. The quantity $I_c = H(B)_{\sigma} - H(E)_{\sigma}$ is called the *coherent information*. Thus, a channel is said to be “not too noisy” if $Q(\mathcal{N}) > 0$. Alternatively, the quantum capacity may also be understood as the ratio of the number of qubits we intend to send to the number of qubits the encoder generates to allow for faithful decoding.

For an arbitrary channel computing $Q(\mathcal{N})$ is intractable. However, a special class of channels known as *degradable* channels admits a single-letter formula $Q(\mathcal{N}) = Q^{(1)}(\mathcal{N})$, which is relatively simpler to calculate. It turns out that the two cases of perfectly reflecting and perfectly absorbing black holes we are interested in can be described by channels with single-letter capacity, and an explicit formula for $Q(\mathcal{N})$ shall be obtained.

C. Hawking Radiation is Featureless

Quantum vacuum fluctuations near the event horizon can produce particle-antiparticle pairs. One particle escapes to infinity as Hawking radiation, while the other falls in, causing the black hole to lose mass. We shall now discuss the channel isometry that captures this idea and show how pure Hawking radiation carries no information at all. The channel isometry $V : A \rightarrow BE$ corresponding to Hawking emission is [1]

$$V(r_{\omega}) = \prod_{\omega} \exp[r_{\omega}(a_k^{\dagger} b_{-k}^{\dagger} - a_k b_{-k})] \quad (2.4)$$

$$V(r_{\omega}) |\text{vac}\rangle = \prod_{\omega} \frac{1}{\cosh r_{\omega}} \sum_{n=0}^{\infty} \tanh^n r_{\omega} |n\rangle_B |n\rangle_E, \quad (2.5)$$

where $\tanh r_{\omega} = e^{-\pi\omega/\kappa}$, $\kappa = 1/2M$ is the surface gravity. In Eq (2.4), a_k annihilates the receiver's vacuum (B) while b_k annihilates vacuum states beyond the black hole horizon (E). Note the resemblance of this isometry to the two-mode squeezing operator from quantum optics, which strongly correlates the two modes and also quantifies our intuition for the mechanism underlying Hawking radiation, as discussed in the beginning of this section. It is well known that the reduced state of either subsystem is thermal: $\varrho_{\omega,B}^{\text{th}} = \varrho_{\omega,E}^{\text{th}}$. This channel is clearly symmetric between Bob and the environment, both have equal amounts of information, and so the quantum capacity vanishes in Eq (2.3). Thus, Hawking radiation alone cannot be used to reconstruct Alice's quantum state.

III. THE BLACK HOLE AS A QUANTUM CHANNEL

A. Sorkin's Late-Time Formulation

The black hole thermalizes an input vacuum state, but what if the input was a generic number state? How does

a black hole interact with Alice's quantum information thrown in at late-times? To go beyond Hawking's result, Brádrler and Adami use Sorkin's consistent extension of semiclassical gravity [8], which describes the interaction of a late-time massless scalar field with an already-formed black hole via the Bogoliubov transformation

$$a_{\text{out}} = \alpha a - \beta b^\dagger + \gamma c, \quad (3.1)$$

with $\alpha, \beta, \gamma \in \mathbb{R}$ and $\alpha^2 - \beta^2 + \gamma^2 = 1$. Here a annihilates early-time modes outside the horizon, b annihilates early-time modes inside the horizon (the Hawking partner modes, inaccessible to Bob), and c annihilates Alice's *late-time* input mode (Fig. 2). Since the late-time field's support is spectrally distinct from Hawking modes, the two sectors commute and can be studied independently.

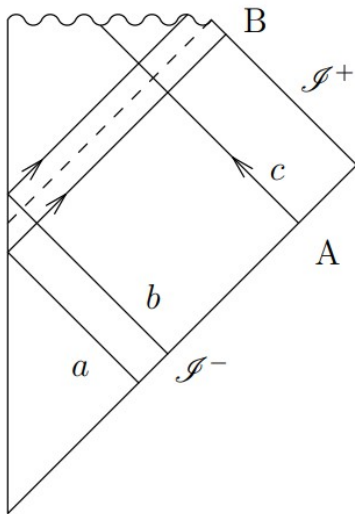


FIG. 2: The dashed line demarcates the horizon and the wavy boundary is the singularity. The interaction of the late-time incoming mode with the horizon is governed by Eq. (3.1). Alice (A) uses the late-time mode c to send her quantum message inside the black hole. Bob (B) collects the radiation at $\mathcal{I}^+$. Adapted from Ref. [1].

This formulation reproduces Hawking's grey-body factors in a thermodynamically consistent manner [3], and correctly predicts superradiance for non-rotating black holes [9]. The key insight is that stimulated emission, is the physical mechanism by which information escapes, and not through Hawking radiation. The dynamics of stimulated emission gives rise to the *Unruh channel*.

B. The Unruh Channel Connection

The Unruh channel describes quantum information transmission between an inertial sender and a uniformly accelerating receiver. In Rindler coordinates, the Bogoliubov transformation relating Minkowski modes d_Ω to Rindler modes $b_\Omega^{R/L}$ is identical in structure to Eq. (3.1).

The channel isometry acts as

$$V_\Omega^{A \rightarrow BE}(|n\rangle_A) = \frac{1}{\cosh^{1+n} r_\Omega} \sum_{m=0}^{\infty} \sqrt{\binom{n+m}{n}} \tanh^m r_\Omega \times |n+m\rangle_B |m\rangle_E \quad (3.2)$$

where $\tanh r_\Omega = e^{-\pi\Omega}$. But since particle number states alone cannot be considered qubits, we switch to considering a dual-rail encoding of a qubit in $\mathbb{C}^2$ spanned by $\{|01\rangle_{A_1 A_2}, |10\rangle_{A_1 A_2}\}$. The isometry then becomes

$$V_\omega^{A_1 \rightarrow B_1 E_1} \otimes V_\omega^{A_2 \rightarrow B_2 E_2} = V_\omega^{A_{12} \rightarrow B_{12} E_{12}}$$

with the qubit Unruh channel being defined as $\mathcal{N} := \text{Tr}_{E_{12}}[V_\omega^{A_{12} \rightarrow B_{12} E_{12}}]$. The resulting qubit channel decomposes as an orthogonal convex sum (direct-sum channel) of *optimal quantum cloning channels*:

$$\mathcal{N} = \bigoplus_{\ell=1}^{\infty} p_\ell \mathcal{Cl}_{1,\ell}, \quad p_\ell = \frac{1}{2}(1-z)^3 \ell(\ell+1) z^{\ell-1}, \quad (3.3)$$

where $z = \tanh^2 r_\Omega$ and $\mathcal{Cl}_{1,\ell}$ is the optimal $1 \rightarrow \ell$ cloning channel [13]. Note that such a channel does not literally clone a state, since that would violate the celebrated no-cloning theorem of quantum mechanics. Instead, it produces an approximation of the input state, with accuracy measured by the fidelity. An *anti-clone* is defined to be the complex conjugate of a cloned state. Therefore, a cloning channel $\mathcal{Cl}_{n,\ell}$ takes n inputs and produces n approximate clones and $\ell - n$ anti-clones.

Cloning channels $\mathcal{Cl}_{1,\ell}$ are also examples of *degradable* channels [14], and have single-letter quantum capacity. It can be shown that $Q(\mathcal{Cl}_{1,\ell}) = \log_2 \frac{\ell+1}{\ell}$ and so from Eq. (3.3) we have

$$Q(\mathcal{N}) = \frac{(1-z)^3}{2} \sum_{\ell=1}^{\infty} \ell(\ell+1) z^{\ell-1} \log_2 \frac{\ell+1}{\ell}. \quad (3.4)$$

where $z = \tanh^2 r_\Omega$ and we have used Lemma 1 of [1], which proves that the capacity of a probabilistic mixture of channels is the probabilistic mixture of capacities.

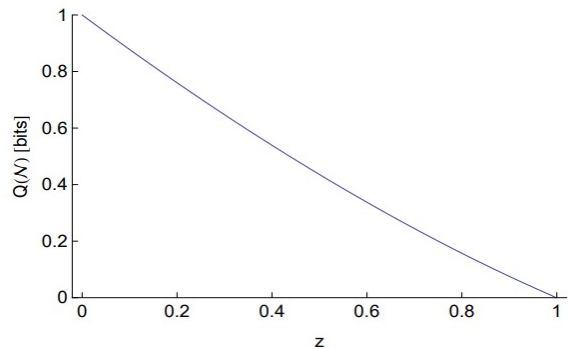


FIG. 3: The quantum capacity of the black hole channel Eq. (3.3) where $z = \tanh^2 r_\omega$. Adapted from Ref. [1].

IV. TWO LIMITING CASES

A. Perfectly Reflecting Black Hole ($\alpha = 0$)

Setting $\alpha = 0$ in Eq. (3.1) eliminates the early-time exterior mode and ensures that the output mode a_{out} depends only on Alice's input in mode c . A comparison shows that the resulting transformation is *exactly* the Unruh isometry Eq. (3.2) under the identification $\kappa \leftrightarrow a$ (surface gravity $\leftrightarrow$ proper acceleration). The two channels are identical, and so we can immediately use all the known results for the Unruh channel from Eq. (3.3) in this limit.

The variation of $Q(\mathcal{N})$ with z is shown in Fig. 3. The capacity is strictly positive for all $z \in [0, 1)$ and since $\kappa = -2\pi\omega/\log_2 z$, the plot tells us that for any macroscopic black hole (any finite surface gravity) Bob can, in principle, perfectly reconstruct Alice's quantum state from the stimulated emission, provided sufficiently many copies of the channel are available for quantum error correction. The capacity vanishes as $z \rightarrow 1$ or $\kappa \rightarrow \infty$, but it should also be noted that this is the limit of microscopic black holes in their final stages of evaporation where the semiclassical description breaks down.

The complementary channel to the Unruh channel $\hat{\mathcal{N}}$ is obtained by tracing out Bob's system instead, and routes information into the black hole interior. It is a direct sum of *anti-clone* channels $\hat{\mathcal{C}}_{1,\ell}$ and it was shown in [13] that it is *entanglement-breaking*: for every input state φ_{RA} , the output $(\text{id}_R \otimes \hat{\mathcal{N}})(\varphi_{RA})$ is separable. [21] Entanglement-breaking channels have zero quantum capacity [23] [24], so $Q(\hat{\mathcal{N}}) = 0$.

Therefore, the interior receives only *anti-clones* of Alice's state, which are insufficient to reconstruct her quantum information. Meanwhile Bob, on the exterior, receives imperfect *clones* that *can* be used (via error correction) to reconstruct her state. The no-cloning theorem is manifestly respected because exact cloning is forbidden; what is transmitted are optimal *approximate* clones, and only one party at a time can access sufficient information to reconstruct the original.

B. Perfectly Absorbing Black Hole ($\alpha = 1$)

In the opposite extreme, the Bogoliubov transformation is implemented by the Hamiltonian

$$H_S = ig_\omega(a^\dagger b^\dagger - ab + a^\dagger c - ac^\dagger), \quad (4.1)$$

giving $a_{out} = e^{-iH_S} a e^{iH_S} = a - g_\omega(b^\dagger + c)$ with $\gamma^2 = \beta^2 \equiv g_\omega^2$ in Eq. (3.1). The interaction between c - and a -modes in Eq. (4.1) takes the form of a beamsplitter, while the a - b coupling produces two-mode squeezing. The input quantum information will be dual-rail encoded in the late-time mode c and the output of the black hole channel will be collected by an outside observer in mode a_{out} .

As earlier, we define the dual-rail isometry as

$$V_\omega^{A_1 \rightarrow B_1 E_1} \otimes V_\omega^{A_2 \rightarrow B_2 E_2} = V_\omega^{A_{12} \rightarrow B_{12} E_{12}}$$

where the channel input system A_{12} denotes the c mode and carries the late-time quantum message encoded as a dual-rail qubit, the channel output by B_{12} corresponds to the a mode and the channel complementary output E_{12} is the b and c mode.

The resulting exterior channel from Alice to Bob after tracing out the environment is given as

$$\mathcal{M} = \bigoplus_{\ell=1}^{\infty} p_\ell \mathcal{D}_\ell, \quad (4.2)$$

for some probability distribution p_ℓ where the first output is a qubit depolarising channel $\mathcal{D}_1(\varrho) = (1-q)\varrho + \frac{q}{2}\text{id}_2$ with $q = 2/3$. This can be verified to be entanglement-breaking via the *positive partial transpose* (PPT) criterion [21] as follows: a channel Φ is entanglement-breaking iff its Choi state $J(\Phi) = (\text{id} \otimes \Phi)(|\Phi^+\rangle\langle\Phi^+|)$ is separable. [22] The Choi state of $\mathcal{D}_1$ is the isotropic state $J(\mathcal{D}_1) = (1-q)|\Phi^+\rangle\langle\Phi^+| + \frac{q}{4}\mathbf{I}_4$. Taking the partial transpose over Bob's subsystem and decomposing in the eigenspaces of the swap operator F (eigenvalues ± 1) gives $J(\mathcal{D}_1)^{T_B} = \frac{1-q}{2}F + \frac{q}{4}\mathbf{I}_4$, whose minimum eigenvalue (from the antisymmetric sector) is

$$\lambda_{\min} = -\frac{1-q}{2} + \frac{q}{4} = \frac{3q-2}{4}. \quad (4.3)$$

For separability we require $\lambda_{\min} \geq 0$, i.e. $q \geq \frac{2}{3}$, so at $q = \frac{2}{3}$ the Choi state is indeed separable, and so $\mathcal{D}_1$ is entanglement-breaking. The higher blocks $\mathcal{D}_\ell$ share the same structure and are also entanglement-breaking [13], hence

$$Q(\mathcal{M}) = 0. \quad (4.4)$$

So Bob receives Alice's imperfect clones but he cannot reconstruct the entanglement. Information is lost, but this does not violate any quantum physics. By contrast, the complementary channel $\hat{\mathcal{M}}$ transmits information to the black hole interior. Now there are *two* copies of the quantum state behind the horizon (a clone and an anti-clone). The interior channel is not entanglement-breaking and has $Q(\hat{\mathcal{M}}) > 0$: the interior can, in principle, reconstruct Alice's state. This is precisely complementary to the reflecting case.

C. Summary of Results

The two cases together paint a consistent and elegant picture, summarised in Table I. In both extremes, $Q(\mathcal{N}_{\text{ext}})$ and $Q(\hat{\mathcal{N}}_{\text{int}})$ cannot both be non-zero simultaneously. This is not a coincidence: it is a consequence of the quantum no-cloning theorem. If both exterior and interior channels had positive quantum capacity, Bob and the interior would each independently hold a perfect reconstruction of Alice's state, violating the no-cloning theorem. The channel-theoretic calculation automatically enforces this constraint.

	$\alpha = 0$ (reflecting)	$\alpha = 1$ (absorbing)
$Q(\mathcal{N}_{\text{ext}})$	> 0	$= 0$
$Q(\mathcal{N}_{\text{int}})$	$= 0$	> 0
Exterior channel type	Cloning (degradable)	Depolarising (EB)
Interior channel type	Anti-cloning (EB)	Clone + anti-clone
No-cloning violated?	No	No

TABLE I: Summary of quantum capacities for the two limiting cases. EB = entanglement-breaking.

Moreover, because the findings hold within standard semiclassical quantum mechanics with no new assumptions, the authors argue that the loss of information posited as part of the information “paradox” is entirely natural within the framework of quantum Shannon theory.

V. CONCLUSION

A. Contextualising the Problem Today

A key limitation of the present work is that quantum capacities are computed only in the perfectly reflecting and perfectly absorbing limits, where single-letter capacity formulae exist. For intermediate reflectivities $0 < \alpha < 1$, the channel is a bosonic Gaussian channel [15] and its quantum capacity remains largely unknown, representing a natural next target.

The information paradox has also been approached by proposing the idea of *black hole complementarity* [25], which posits that an observer beyond the event horizon and an observer outside each see consistent pictures complementary to each other. Even if quantum information is cloned due to reflection and transmission across the boundary, any observable physics is not violated due to the inaccessibility of the inside by the outside and vice versa. This was challenged later by AMPS [19] who proposed the *firewall paradox*, which argued that unitarity, the equivalence principle, and effective field theory cannot simultaneously hold at the horizon unless we violate the principle of the monogamy of entanglement. Further, the AdS/CFT correspondence [26] between the bulk of a gravitational system and its quantum boundary tells

us that dual to every event in the bulk is a quantum description of dynamics on the boundary. This implies that the formation and evaporation of black holes have an equivalent description in quantum dynamics, where the information cannot be lost thus implying black holes do not destroy information. Though the claim is strong, the exact mechanism underlying the escape of information behind an event horizon has not been understood yet.

Black holes are conjectured to be nature’s fastest scramblers [16], thermalising information in a scrambling time $t_* \sim \beta \log S$, with β being the inverse temperature and S denoting entropy. This scrambling is precisely what Hayden and Preskill [4] exploit to argue that information escapes shortly after the Page time. The *holographic duality* underlying this picture connects the black hole’s boundary quantum system to random matrix theory and quantum chaos: the Sachdev-Ye-Kitaev model [17] saturates the bound on the Lyapunov exponent [18] and provides a concrete, solvable pipeline from black hole physics to quantum chaos. Computing the quantum capacity of the effective channel induced by such maximally scrambling dynamics would directly bridge the channel-theoretic approach reviewed here with the broader programme of quantum gravity through quantum information.

B. Final Remarks

We have reviewed the information paradox in the context of the work by Brádler and Adami [1], which treats a semiclassical Schwarzschild black hole as a quantum channel and computes its quantum capacity in two exactly solvable limits. The main findings are: (i) Hawking radiation alone carries zero quantum information; (ii) in the perfectly reflecting limit, the black hole channel is identical to the Unruh channel, and has a non-vanishing quantum capacity, (iii) the complementary interior channel is entanglement-breaking with zero capacity, ensuring the no-cloning theorem is respected; (iv) in the perfectly absorbing limit the roles are exactly reversed. In neither case can both the interior and exterior simultaneously reconstruct Alice’s quantum state, in perfect accord with the no-cloning theorem. The black hole information paradox in these limiting cases is then not a paradox at all; it is simply quantum Shannon theory at work.

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